

ZIHAO ZHU (朱梓豪)

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EDUCATION

The Chinese University of Hong Kong, Shenzhen	<i>2021.09 - Present</i>
School of Data Science	
Ph.D. of Data Science	
Advisor: Baoyuan Wu	
University of Chinese Academy of Sciences	<i>2018.09 - 2021.06</i>
Institute of Information Engineering	
Master of Science in Engineering	
Advisor: Yue Hu, Jing Yu	
China University of Mining and Technology	<i>2014.09 - 2018.06</i>
School of Computer Science and Technology	
Bachelor of Engineering	

RESEARCH INTERESTS

AI security & privacy, Multimodal large language models

RESEARCH EXPERIENCE

Tencent AI Lab	Shenzhen, China
Research Intern	<i>2024.03 - Present</i>
Shenzhen Research Institute of Big Data	Shenzhen, China
Research Assistant at Secure Computing Lab of Big Data	<i>2021.06 - 2024.01</i>

PUBLICATIONS

Journal (1 PR)

1. [Cross-Modal Knowledge Reasoning for Knowledge-based Visual Question Answering](#)
Jing Yu*, **Zihao Zhu**, Yujing Wang, Weifeng Zhang, Yue Hu, Jianlong Tan (*Advisor)
In Pattern Recognition (PR), 2020

Conference (1 ICLR, 1 NeruIPS, 2 IJCAI, 1 AAAI, 2 ICASSP)

1. [VDC: Versatile Data Cleanser for Detecting Dirty Samples via Visual-Linguistic Inconsistency](#)
Zihao Zhu, Mindda Zhang, Shaokui Wei, Bingzhe Wu, Baoyuan Wu
In International Conference on Learning Representations (ICLR), 2024
2. [Learning to Optimize Permutation Flow Shop Scheduling via Graph-based Imitation Learning](#)
Longkang Li, Siyuan Liang, **Zihao Zhu**, Xiaochun Cao, Chris Ding, Hongyuan Zha, Baoyuan Wu
In AAAI Conference on Artificial Intelligence (AAAI), 2024

3. [BackdoorBench: A Comprehensive Benchmark of Backdoor Learning](#)
Baoyuan Wu, Hongrui Chen, Mingda Zhang, **Zihao Zhu**, Shaokui Wei, Danni Yuan, Hongyuan Zha
In Thirty-sixth Conference on Neural Information Processing Systems (NeurIPS), 2022
4. [From Shallow to Deep: Compositional Reasoning over Graphs for Visual Question Answering](#)
Zihao Zhu
In IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), 2022
5. [MCR-NET: A Multi-Step Co-Interactive Relation Network for Unanswerable Questions on Machine Reading Comprehension](#)
Wei Peng, Yue Hu, Luxi Xing, Yuqiang Xie, **Zihao Zhu**, Yajing Sun
In IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), 2021
6. [Mucko: Multi-Layer Cross-Modal Knowledge Reasoning for Fact-based Visual Question Answering](#)
Zihao Zhu, Jing Yu, Yujing Wang, Yajing Sun, Yue Hu, Qi Wu
In Proceedings of the 29th International Conference on Joint Artificial Intelligence (IJCAI), 2020
7. [DAM: Deliberation, Abandon and Memory Networks for Generating Detailed and Non-repetitive Responses](#)
Xiaoze Jiang, Jing Yu, Zengchang Qin, **Zihao Zhu**, Qi Wu
In Proceedings of the 29th International Conference on Joint Artificial Intelligence (IJCAI), 2020

Preprints (5 arXiv)

1. [Boosting Backdoor Attack with A Learnable Poisoning Sample Selection Strategy](#)
Zihao Zhu, Mingda Zhang, Shaokui Wei, Li Shen, Yanbo Fan, Baoyuan Wu
arXiv, 2023
2. [BlackboxBench: A Comprehensive Benchmark of Black-box Adversarial Attacks](#)
Meixi Zheng, Xuanchen Yan, **Zihao Zhu**, Hongrui Chen, Baoyuan Wu
arXiv, 2023
3. [Versatile Backdoor Attack with Visible, Semantic, Sample-Specific, and Compatible Triggers](#)
Ruotong Wang, Hongrui Chen, **Zihao Zhu**, Li Liu, Baoyuan Wu
arXiv, 2024
4. [Attacks in Adversarial Machine Learning: A Systematic Survey from the Life-cycle Perspective](#)
Baoyuan Wu, **Zihao Zhu**, Li Liu, Yanbo Fan, Siwei Lyu, Hongyuan Zha
arXiv, 2023
5. [Defenses in Adversarial Machine Learning: A Survey](#)
Baoyuan Wu, Shaokui Wei, Mingli Zhu, Meixi Zheng, **Zihao Zhu**, Mingda Zhang, Hongrui Chen, Danni Yuan, Li Liu, Qingshan Liu
arXiv, 2023

SERVICE

Reviewer:

AAAI 2020, NeurIPS 2023, CVPR 2024, NeurIPS 2024

TEACHING

MFE5260 Data Sciences in Financial Engineering
Teaching Assistant

2022 Spring

DDA6309 Probabilistic Graphical Models
Teaching Assistant

2021 Fall

AWARDS

AIRS Talent of PhD Research Program

2021.09

University of Chinese Academy of Sciences Merit Student

2020.10

Outstanding Undergraduate Thesis Award

2018.06

The First Prize Scholarship of CUMT

2016.10

The First Prize Scholarship of CUMT

2015.10